

ISE 331 - Fundamentals of Computer Security
Homework Assignment 04

Assigned: Tuesday, 03/24/2026

Due: Tuesday, 04/07/2026, by 11:59 PM

Points: 30 points (5 pts. per question)

Please complete and turn in responses to the following questions from the textbook:

Chapter 09, Essay Questions: 1, 3, 4, and 5

9.1. A developer in your company is building a new application and has asked you if she should use TCP- or UDP-based communications. Provide her with a brief discussion of the advantages and disadvantages of each protocol.

9.3. Describe three basic types of network topologies and provide a sample diagram of each type.

9.4. Describe the three-way handshake process used to initiate TCP connections.

9.5. Your boss wants to know how subnetting works. Provide her with a brief description and be sure to include an example to illustrate how subnetting works.

Chapter 17, Essay Questions: CMK.17.1, CMK.17.2

CMK.17.1. Explain how a TLS connection is established.

CMK.17.2. What is the purpose of the random nonces in the SSL/TLS handshake?

You are encouraged to also complete the "Key Terms Quiz" and the "Multiple Choice Quiz" for chapters 9 and 17, but you are not required to turn those in.

I have posted copies of the End of Chapter exercises from the textbook for chapters 9 and 17 along with this assignment.

Submission instructions:

1. Please write your completed assignment in a word processor or text editor.
 - No scans of handwritten prose will be accepted.
 - Scans of hand-drawn diagrams are acceptable.

2. Please submit your completed assignment through Blackboard in PDF format.
3. Please name your file ise331_hw4_LastnameFirstname.pdf
3. At the top of your submission, please include the following:
 - Name
 - SBU ID Number
 - SBU netid
 - Date
 - ISE 331, HW 4
4. Exam questions will be based on the homework assignments. Taking the homework assignments seriously is a good way to prepare for the exams.
5. These are individual assignments. Any collaboration on answers will be considered a violation of academic honesty.